

Notes: Hastings, Hortaçsu, and Syverson (Econometrica, 2017)

Sales Force and Competition in Financial Product Markets: The Case of Mexico's Social Security Privatization

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Institutional setting and data	4
3.1	Mexico's privatized social security system	4
3.2	Afore fees	4
3.3	Administrative data	4
3.4	Sales force exposure	5
3.5	Price sensitivity and brand value	5
4	Model and empirical specifications	5
4.1	Benchmark indirect utility	5
4.2	Choice probabilities with sales force	5
4.3	Empirical logit specification	6
4.4	Sales force and brand value	6
4.5	Sales force and price sensitivity	6
4.6	Expected cost and counterfactuals	6
5	Identification and empirical design	7
5.1	What identifies the effect of sales force	7
5.2	Why endogeneity is a concern	7
5.3	Instrumental variables strategy	7
5.4	Why demographic heterogeneity matters	7
5.5	What remains imperfect	8
6	Tables and figures	8
6.1	Figure 1 and Figure 2: sales force and fees over time	8
6.2	Table I and Table II: descriptive statistics and weak price sensitivity	8
6.3	Figure 3 and Figure 4: sales force, markups, and market share	9
6.4	Table III and Table IV: sales force and Afore brand value	10
6.5	Table V and Figure 5: sales force reduces price sensitivity	10
6.6	Table VI: neutral advertising and consumer costs	11
6.7	Tables VII–IX: policy counterfactuals	11
7	Short summary	12
8	Conclusion	13
8.1	Core conclusion	13
8.2	Economic intuition	13

8.3	Incremental contribution takeaway	13
8.4	Policy takeaway	13

1 Big picture

- **Research question.** How does sales-force-based marketing affect competition, equilibrium prices, and consumer welfare in a privatized financial product market?
- **Setting.** The paper studies the 1997 launch of Mexico’s privatized social security system. Workers had to choose an account manager, called an Afore, from a set of regulated firms. The products were highly similar, but fees were high and sales forces were huge.
- **Central empirical fact.** The Afores with the largest sales forces also charged high fees and obtained large market shares. Consumers did not generally choose the lowest-cost manager for their own account profile.
- **Core mechanism.** Sales agents can have two effects. They can increase the salience or brand value of the firm they represent; and they can reduce effective price sensitivity, making consumers less responsive to management costs.
- **Demand model.** The authors estimate a model of Afore choice in which utility depends on expected account management costs, Afore brand value, demographics, and local exposure to sales agents.
- **Identification strategy.** The key variation is local geographic variation in sales-force exposure across municipalities and demographic cells. The authors exploit the spatial distribution of Afores’ sales agents, together with instruments based on local bank-branch networks and competitor sales-force patterns.
- **Main estimate.** Local sales force exposure raises the perceived brand value of an Afore and lowers price sensitivity. This means sales force is not merely informative; it can make consumers less responsive to fees.
- **Counterfactual result.** If sales agents had no effect on preferences, demand elasticities would be much higher and total management costs would fall materially.
- **Policy result.** A government competitor alone is not enough to restore competition when many consumers are highly inelastic. Policies that increase price sensitivity or reduce persuasive sales effects are necessary.
- **Economic takeaway.** Competition in social-safety-net financial markets can fail when firms compete through marketing that weakens price sensitivity. Demand-side consumer protection can be as important as supply-side competition.

2 Incremental contribution of the paper

- **Relative to descriptive work on high fees.** The paper does not merely document that fees were high in privatized pension accounts. It provides a structural demand-and-counterfactual framework that explains why high fees were sustainable.
- **Relative to classic advertising models.** Traditional informative advertising models predict that marketing can expand awareness and strengthen competition. This paper shows that sales-force marketing in financial products can instead lower price sensitivity and soften competition.
- **Relative to industrial organization studies of financial products.** The paper brings rich administrative individual-level data to a setting where firms set national prices but choose local sales effort. This separation is especially useful for identifying the role of nonprice competition.
- **Relative to behavioral household finance.** The paper links low financial sophistication and consumer inattention to equilibrium market power. It shows that firms’ marketing choices are not just responses to consumer mistakes; they can amplify those mistakes.
- **Relative to policy debates on privatized pensions.** The paper quantifies why adding a low-cost public option may be insufficient. If consumers remain price-inelastic, private firms may not reduce fees enough. Demand-side interventions that increase price sensitivity are needed.

- **Methodological increment.** The model allows sales force to affect both brand value and the marginal utility of price. This is more general than treating advertising as only a demand shifter.
- **Broader contribution.** The paper provides a template for studying competition in markets where financial intermediaries sell complex, low-salience products through brokers, agents, or sales representatives.

3 Institutional setting and data

3.1 Mexico’s privatized social security system

- Mexico launched a fully privatized defined-contribution social security system in 1997.
- Workers were required to place retirement contributions into individual accounts managed by private fund administrators known as Afores.
- There were 17 regulator-approved Afores at the system’s launch.
- Investment portfolios were heavily regulated, making products relatively homogeneous on investment attributes.
- Despite this, fees were high and differed substantially across firms.
- Switching was technically possible but difficult in practice because the paperwork and relationship were controlled by the incumbent Afore.

Interpretation: The institutional design creates a useful empirical environment: products are similar, price formulas are observable, firms set fees at a national level, and sales force varies locally.

3.2 Afore fees

Afores charged two main fees:

- a **flow fee**, charged as a fraction of each contribution;
- a **balance fee**, charged as a fraction of the account balance.

For a worker i and Afore j , the paper constructs the expected management cost:

$$C_{ij} = \text{expected present value of fees paid by worker } i \text{ if assigned to Afore } j. \quad (1)$$

Interpretation: The expected cost differs across workers because flow fees matter more for workers with higher future contributions, while balance fees matter more for workers with high balances or long horizons.

3.3 Administrative data

- The data include all workers in the system from 1997 to 2007.
- For each worker, the data include account choice, contributions, balances, wages, gender, date of birth, and zip code of residence for most workers.
- The authors use worker characteristics to project future contributions and balances and therefore compute personalized costs of each Afore.
- Sales-agent data come from CONSAR’s official registry of sales agents, including their Afore affiliation, status, and location.
- The paper also uses local municipality and bank-branch information to construct instruments and controls.

Interpretation: The richness of the data allows the authors to calculate the cost-minimizing Afore for each worker, rather than using a single aggregate fee ranking.

3.4 Sales force exposure

The key local sales-force measure is a municipality-level concentration:

$$SF_{jm} = \frac{\text{number of sales agents for Afore } j \text{ in municipality } m}{\text{number of SAR affiliates in municipality } m}. \quad (2)$$

The paper also uses total sales force exposure and bank-branch concentration variables.

Interpretation: Since fees are set nationally while sales agents are deployed locally, local variation in sales force is informative about the effect of sales effort on demand.

3.5 Price sensitivity and brand value

The demand model estimates:

- a cell-specific cost sensitivity parameter λ_c ;
- an Afore-by-cell brand valuation δ_{cj} ;
- the dependence of both on local sales-force exposure.

Interpretation: A sales agent can raise demand by increasing δ_{cj} , or by making λ_c closer to zero, which means the consumer becomes less responsive to management costs.

4 Model and empirical specifications

4.1 Benchmark indirect utility

The stylized benchmark model starts with client i buying from Afore j through agent A_j :

$$u_{iA_j} = \lambda_i C_{ij} + \delta_j + \varepsilon_{iA_j}. \quad (3)$$

Here:

- C_{ij} is the expected management cost;
- λ_i is the marginal utility of cost, expected to be negative;
- δ_j is the nonprice brand value of Afore j ;
- ε_{iA_j} is a match-specific utility shock.

Interpretation: A lower-cost Afore should be chosen more often if λ_i is substantially negative. A large positive δ_j can offset high costs.

4.2 Choice probabilities with sales force

If Afore j has N_j sales agents in the market, the probability of choosing Afore j is:

$$\Pr(i \text{ chooses } j) = \frac{N_j \exp(\lambda_i C_{ij} + \delta_j)}{\sum_k N_k \exp(\lambda_i C_{ik} + \delta_k)}. \quad (4)$$

Equivalently,

$$\Pr(i \text{ chooses } j) = \frac{\exp(\lambda_i C_{ij} + \delta_j + \ln N_j)}{\sum_k \exp(\lambda_i C_{ik} + \delta_k + \ln N_k)}. \quad (5)$$

Interpretation: Sales force enters like a demand shifter because more agents create more opportunities for a worker to encounter and sign with that Afore.

4.3 Empirical logit specification

At the demographic-geographic cell level, the choice model can be summarized as:

$$u_{icj} = \lambda_c C_{ij} + \delta_{cj} + \varepsilon_{icj}. \quad (6)$$

The authors estimate cell-level cost sensitivity and Afore-specific brand values. The flexible cell structure captures differences by geography and demographics.

Interpretation: This specification lets low-income, high-income, young, old, male, and female workers differ in how they respond to cost and nonprice Afore attributes.

4.4 Sales force and brand value

The first key second-stage regression is:

$$\delta_{cj} = a + b SF_{jm} + \Gamma X_{jm} + \eta_{cj}. \quad (7)$$

The paper instruments sales force exposure using local variation in bank-branch networks and other local determinants of sales-agent placement.

Interpretation: If $b > 0$, sales force raises brand value. This is consistent with persuasive marketing, salience, convenience, or relationship-based selling.

4.5 Sales force and price sensitivity

The second key regression studies whether total sales-force exposure changes the cost sensitivity parameter:

$$\lambda_c = a + \gamma TotalSF_c + \Pi Z_c + \nu_c. \quad (8)$$

Interpretation: If $\gamma > 0$ and $\lambda_c < 0$, then sales force makes the absolute value of λ_c smaller. That is, sales force reduces price sensitivity.

4.6 Expected cost and counterfactuals

For a worker with preferences θ_i , expected cost is:

$$E(C_i | \theta_i) = \sum_j C_{ij} q_j(\theta_i), \quad (9)$$

where $q_j(\theta_i)$ is the model-implied probability of choosing Afore j .

Counterfactuals set the effect of sales force on preferences to zero, add a government competitor, or increase price sensitivity for the most inelastic consumers.

Interpretation: The counterfactuals quantify how much consumer welfare and fees change if sales force no longer rotates demand away from low-cost options.

5 Identification and empirical design

5.1 What identifies the effect of sales force

- Fees are set nationally, but sales force varies locally.
- Workers in different municipalities face the same menu of prices but different exposure to Afore agents.
- This separates national price policy from local nonprice competition.
- The authors control for local demographics and Afore fixed effects, and use instruments for local sales-force placement.

Interpretation: The key idea is to compare how Afores perform across locations where their sales-force intensity differs while their national fees are the same.

5.2 Why endogeneity is a concern

Sales force placement could be endogenous if Afores send agents to locations where they expect stronger demand. For example:

- Santander may send more agents to municipalities where workers already like Santander;
- high-income areas may attract more agents and also have different fee sensitivity;
- bank affiliations may correlate with financial familiarity.

Interpretation: Without instruments and controls, positive sales-force effects could reflect latent demand rather than causal influence.

5.3 Instrumental variables strategy

The paper uses instruments based on local bank-branch structure and related local variation. Bank-branch concentration affects the feasibility and familiarity of sales-agent operations but is argued to be excluded from Afore choice after controlling for relevant observables and fixed effects.

Interpretation: The IV strategy attempts to isolate exogenous variation in sales-agent exposure from endogenous placement into attractive customer markets.

5.4 Why demographic heterogeneity matters

The paper estimates effects separately for:

- low-wage and high-wage workers;
- men and women;
- younger and older workers;
- cells where an Afore is one of the lowest-cost options.

Interpretation: If sales force is persuasive, effects may be stronger among groups with lower financial sophistication or greater responsiveness to face-to-face marketing.

5.5 What remains imperfect

- Sales-agent contacts are not observed at the individual level.
- Instruments rely on exclusion restrictions that cannot be tested directly.
- The model is static because switching was nearly absent in the early system.
- Counterfactuals hold some cost and conduct assumptions fixed.

Interpretation: The empirical design is strong because of detailed administrative data and local variation, but it still depends on structural and IV assumptions.

6 Tables and figures

6.1 Figure 1 and Figure 2: sales force and fees over time

Read:

- Figure 1 shows that sales force was enormous at the system's launch, peaking around the inception period and then declining substantially.
- The largest Afore sales force closely follows the aggregate pattern, showing intense early competition for account enrollment.
- Figure 2 shows that average and median flow fees were high and relatively stable, while balance fees increased for some firms.
- Fees did not collapse despite many firms in the market and homogeneous regulated products.

Economic message: Early competition was mainly sales-force competition, not price competition. Firms fought for initial account capture and then maintained high fees.

re 1 plots the level of agents in the market over time; the average as well as the largest across Afores. Both statistics decline substantially after the inception period

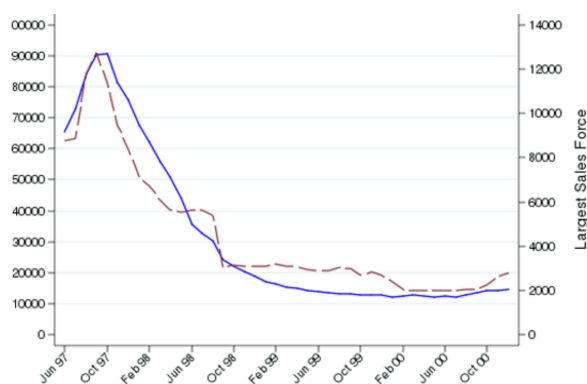


Figure 1: distribution of sales force over time

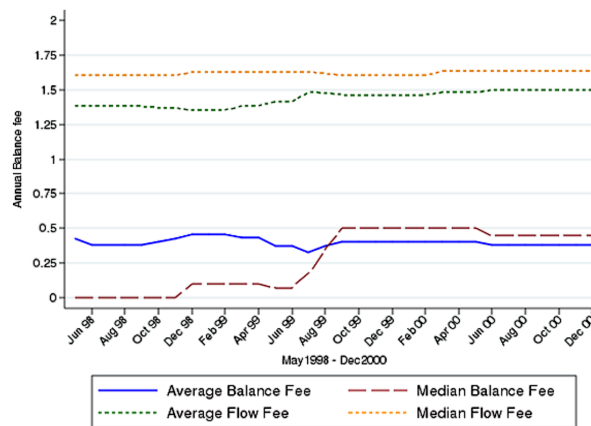


Figure 2: average and median fees

6.2 Table I and Table II: descriptive statistics and weak price sensitivity

Read:

- Table I shows that the highest-share Afores were often expensive and had very large sales forces.
- Santander, Banamex, Bancomer, Profuturo GNP, and Garante had large market shares despite high implied loads.
- The correlation between sales agents and market share is high.

- Table II shows that many workers chose Afores that were not close to the cheapest option for them.
- Markups over the cheapest option are positive for most large-share Afores.

Economic message: The raw data already point to a failure of price competition: high-fee providers win clients when they have high brand value and large sales forces.

TABLE I
DESCRIPTIVE STATISTICS OF AFORES AT THE INCEPTION OF MARKET^a

Afore	Flow Fee	Implied Load on Contributions	Balance Fee	Share of Accounts	Number of Agents
Santander	1.70%	26.15%	1.00%	14.60%	12,361
Garante	1.68%	25.85%	0.00%	9.83%	11,756
Bancreeer/Dresdner/HSBC	0.00%	0.00%	4.75%	4.62%	8,804
Bancomer	1.70%	26.15%	0.00%	16.12%	7,583
Profuturo GNP	1.70%	26.15%	0.50%	12.45%	7,443
Banorte Generali	1.00%	15.38%	1.50%	8.35%	7,440
ING/Bital	1.68%	25.85%	0.00%	9.21%	7,369
Banamex	1.70%	26.15%	0.00%	12.94%	5,914
Previnter	1.55%	23.85%	0.00%	2.28%	4,614
Inbursa	0.00%	0.00%	1.57%	2.52%	4,150
Tepeyac	1.17%	18.00%	1.00%	0.52%	3,685
Genesis Metropolitan	1.65%	25.38%	0.00%	0.94%	3,213
XXI	1.50%	23.08%	0.99%	2.88%	2,521
Atlantico Promex	1.40%	21.54%	0.95%	1.32%	2,045
Principal	0.90%	13.85%	1.00%	1.01%	1,732
Capitaliza	1.60%	24.62%	0.00%	0.23%	925
Zurich	0.95%	14.62%	1.25%	0.18%	910
Total				100%	92,465

^a"Implied Load on Contributions" is the flow fee divided by 6.5% (the share of salary automatically placed in the SAR account). Share of accounts is calculated using all account holders as of June 2006 who entered the SAR 1997 system before June 1998. Santander is a Spanish financial group. Garante is a Mexican insurance and financial group. Bancreeer/Dresdner/HSBC is an international financial group. Bancomer is the second largest Mexican bank. Profuturo GNP is a Mexican insurance group. Banorte Generali is a joint venture between a large northern Mexican bank, Banorte, and the largest Italian insurance company, Assicurazioni Generali S.p.A. ING/Bital is an international financial group. Banamex is the largest Mexican bank. Previnter is a France-based international insurance company, acquired by Profuturo GNP in late 1998. Inbursa is the financial arm of Telcel magnate's Slim Corporation. Tepeyac is a Mexican insurance company. Genesis Metropolitan is owned by the United States-based insurance company, Metropolitan Life. It was acquired by Santander in late 1998. XXI is the Afore branded by IMSS, the former pension system administrator. Atlantico

Table I: descriptive statistics at market inception

TABLE II
DESCRIPTIVE STATISTICS OF MANAGEMENT COSTS, MARKET SHARES, SALES FORCE, AND PRICE SENSITIVITY, BY AFORE^a

	Number of Agents	Share of Accounts	Mean Rank for Own Clients	Mean Rank in the System	Markup Over Cheapest Option	Mean Savings in Days of Wages	Mean Daily Wage of Clients (1997 Pesos)	Fraction of Clients Who Are Male
Santander	12,361	14.6%	16.5	16.5	1.9	42.5	61.2	0.72
Garante	11,756	9.8%	11.1	11.0	1.3	30.7	75.9	0.69
Bancreeer/Dresdner/HSBC	8,804	4.6%	14.2	14.1	1.9	55.7	69.9	0.69
Bancomer	7,583	16.1%	7.4	7.6	1.1	29.4	109.6	0.67
Profuturo GNP	7,443	12.5%	14.4	14.4	1.6	34.6	59.1	0.71
Banorte Generali	7,440	8.4%	8.0	9.2	1.4	34.0	64.3	0.68
ING/Bital	7,369	9.2%	8.1	8.0	1.2	29.0	74.7	0.64
Banamex	5,914	12.9%	9.7	10.0	1.1	28.4	97.8	0.67
Previnter	4,614	2.3%	4.1	4.2	1.0	24.9	96.1	0.65
Inbursa	4,150	2.5%	1.2	1.3	0.0	0.5	218.6	0.65
Tepeyac	3,685	0.5%	7.2	7.6	1.2	27.1	70.1	0.71
Genesis Metropolitan	3,213	0.9%	8.1	7.9	1.2	25.9	63.1	0.64
XXI	2,521	2.9%	14.8	14.8	1.5	39.3	120.7	0.57
Atlantico Promex	2,045	1.3%	13.0	12.9	1.3	38.5	74.6	0.66
Principal	1,732	1.0%	2.4	2.3	0.6	11.4	76.7	0.67
Capitaliza	925	0.2%	6.1	5.5	1.3	23.1	99.4	0.67
Zurich	910	0.2%	5.9	5.5	1.0	26.6	96.8	0.83
Correlation with num. sales agents	1.00	0.79	0.50	0.52	0.48	0.45	-0.28	0.06

^aAfore rank is based on expected costs and runs from 1 to 17, where a 1 indicates the lowest cost. "Mean Rank of Own Clients" is the average expected cost rank of that Afore among the Afore's clients. "Mean Rank in the System" is the average expected cost rank for that Afore over all clients in the system. For each worker, expected costs are calculated by averaging projected costs,

Table II: costs, market shares, sales force, and price sensitivity

6.3 Figure 3 and Figure 4: sales force, markups, and market share

Read:

- Figure 3 plots mean client markup against Afore sales force. Larger sales forces are associated with higher markups.
- Bubble size reflects market share, so the high-sales-force, high-markup firms also attract many accounts.
- Figure 4 shows that municipalities with greater mean sales-force concentration have a stronger positive relation between markup and market share.

Economic message: These figures motivate the structural model. Sales force appears to rotate demand toward higher-cost firms rather than disciplining fees.

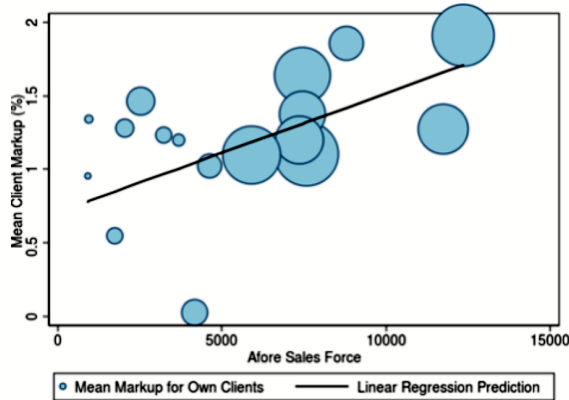


Figure 3: markup, sales force, and market share

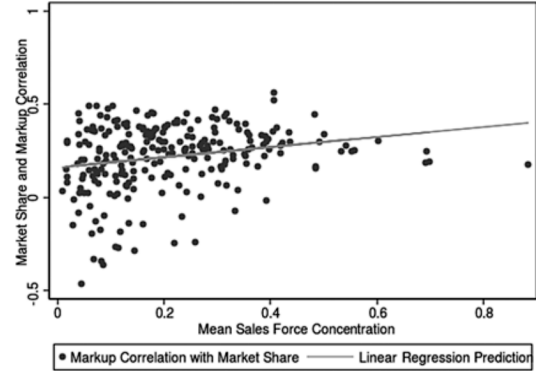


Figure 4: sales force and price-share relation

6.4 Table III and Table IV: sales force and Afore brand value

Read:

- Table III estimates the impact of local sales-force concentration on Afore-specific brand value.
- The OLS coefficient on own sales-force concentration is positive and significant.
- IV estimates are also positive and larger than OLS, suggesting endogenous placement may not fully explain the effect.
- Bank-branch concentration and bank affiliation also increase brand value.
- Table IV shows that the sales-force effect is positive across all demographic groups.
- The effect is larger for low-wage workers than for high-wage workers and for men than for women.

Economic message: Sales agents raise the perceived value of the Afore they represent. This supports a persuasive or salience channel rather than a pure price-information channel.

TABLE III
IMPACT OF SALES FORCE ON AFORE BRAND VALUE^a

Dependent Variable: $\delta_{c,j}$	(1) OLS	(2) IV 1	(3) IV 2	(4) IV 3
Municipality sales force concentration for Afore j	2.793*** (0.020)	4.499*** (0.031)	4.404*** (0.031)	4.169*** (0.029)
Municipality concentration of bank branches for Afore j	28.311*** (0.616)	21.582*** (0.661)	21.957*** (0.657)	22.884*** (0.647)
Indicator if Afore j is affiliated with a bank.	0.715*** (0.012)	0.483*** (0.014)	0.496*** (0.013)	0.528*** (0.013)
Observations	62,883	62,883	62,883	62,883
Mean $\delta_{c,j}$	-2.743			
Mean sales force	0.280			
StDev sales force	0.312			
Mean branches	0.004			
StDev branches	0.010			
F-stat for excluded instruments	—	1,425.71	1,038.04	728.24
Number of cells	3,699	3,699	3,699	3,699

^aNotes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The dependent variable is the estimated mean valuation of Afore j for system affiliates in demographic and municipality cell c estimated using equation (4) in the text. It measures mean value relative to the excluded Afore, whose identity is held constant across all municipalities. All specifications include cell-level fixed effects and an indicator if the Afore had zero market share in that cell. Municipality sales force concentration for Afore j is defined as the number of agents for Afore j in municipality m divided by the total number of SAR affiliates in m . Municipality concentration of bank branches for Afore j is defined as the number of bank branches in municipality m divided by the adult population in the municipality. Column 1 presents OLS estimates. Column 2 presents instrumental variables estimates using the advertising-spillover instrument (mean costs in same-municipality-other-demographic-group cells for Afore j interacted with Afore fixed effects). Column 3 adds competitor bank branches in the municipality and its interaction with Afore fixed effects as instruments. Column 4 adds fraction of the working-age cell population who are IMSS account holders and fraction of workers employed in the public sector, both interacted with Afore fixed effects.

Table III: sales force and Afore brand value

TABLE IV
EFFECT OF SALES AGENTS ON BRAND VALUE^a

Dependent Variable: $\delta_{c,j}$	(1) Low Wage	(2) High Wage	(3) Male	(4) Female	(5) Younger	(6) Older	(7) Low Cost
Municipality sales force concentration for Afore j	4.598*** (0.040)	3.660*** (0.041)	4.650*** (0.040)	3.502*** (0.041)	4.412*** (0.041)	3.908*** (0.040)	4.498*** (0.075)
Municipality concentration of bank branches for Afore j	16.244*** (0.888)	30.302*** (0.938)	20.920*** (0.864)	25.556*** (0.972)	21.553*** (0.915)	24.356*** (0.912)	34.921*** (1.525)
Indicator if Afore j is affiliated with a bank.	0.396*** (0.018)	0.694*** (0.019)	0.507*** (0.017)	0.559*** (0.021)	0.558*** (0.019)	0.496*** (0.019)	0.302*** (0.028)
Observations	34,289	28,594	39,610	23,273	32,385	30,498	14,796
Number of cells	2,017	1,682	2,330	1,369	1,905	1,794	3,699
Mean $\delta_{c,j}$	-2.985	-2.452	-2.779	-2.680	-2.803	-2.679	-3.105

^aInstrumental variables specification from Table III, column 4, by subgroup. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Wage cuts are defined as below or above the median wage. Younger and older are defined as younger or older than 35 years old. "Low Cost" is an indicator if Afore j is one of the cheapest four Afores for individuals in demographic-group-municipality cell c . The dependent variable is the estimated mean valuation of Afore j for system affiliates in demographic and municipality cell c estimated using equation (4). It measures mean value relative to the excluded Afore, whose identity is held constant across all municipalities. All

Table IV: brand value by demographic subgroup

6.5 Table V and Figure 5: sales force reduces price sensitivity

Read:

- Table V regresses estimated mean price sensitivity on total sales-force concentration.
- The coefficient is positive, while the baseline price-sensitivity parameter is negative, so sales force reduces the absolute sensitivity to cost.
- The reduction in price sensitivity is especially strong for low-wage workers, men, and younger workers.
- Figure 5 compares baseline demand elasticities to the counterfactual in which agents have neutral effects.
- Neutralizing agent effects makes demand much more elastic for nearly all Afores.

Economic message: The paper's key mechanism is that sales force both shifts demand toward particular brands and makes consumers less responsive to fees.

Dependent Variable: α_c	(1) Pooled	(2) Pooled	(3) Low Wage	(4) High Wage	(5) Male	(6) Female	(7) Younger	(8) Older
Municipality sales force concentration	0.046*** (0.012)	0.051*** (0.017)	0.060*** (0.020)	0.020*** (0.007)	0.057*** (0.016)	0.019 (0.014)	0.054*** (0.015)	0.035*** (0.013)
Sales force concentration for four lowest-cost Afores		-0.026 (0.045)						
Constant	-0.605*** (0.081)	-0.606*** (0.081)	-0.845*** (0.134)	-0.272*** (0.045)	-0.692*** (0.103)	-0.410*** (0.099)	-0.686*** (0.100)	-0.510*** (0.085)
Observations	3,699	3,699	2,017	1,682	2,330	1,369	1,905	1,794
Mean dep. var.	-0.388	-0.388	-0.568	-0.173	-0.435	-0.308	-0.431	-0.343

^aNotes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Dependent variable is the estimate of mean price sensitivity for each demographic-group-municipality cell. c. from equation (4). Wage cuts are defined as below or above the median wage. Younger and older are

Table V: sales force and price sensitivity

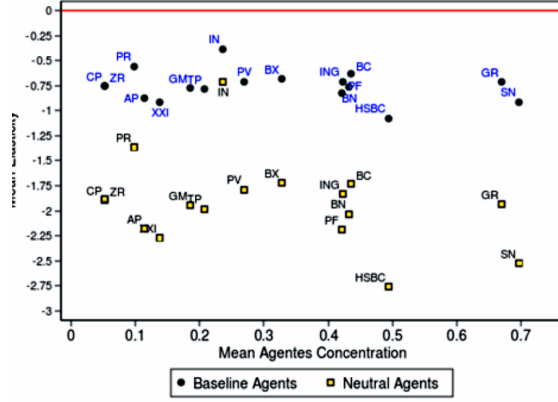


Figure 5: elasticities with baseline vs. neutral agents

6.6 Table VI: neutral advertising and consumer costs

Read:

- Table VI simulates what happens if sales agents have no effect on preferences while observed fees are held fixed.
- Mean elasticity becomes much more negative, from -0.754 to -1.925 overall.
- Total costs fall by 17.2% overall.
- Average savings are 815 pesos per capita, equivalent to about 1.4 months of the minimum pension.
- High-wage and older workers gain more in pesos, but all groups benefit.

Economic message: Sales force materially raises management fees paid by workers. The effect is not just reallocative across firms; it creates large consumer cost losses.

TABLE VI
SIMULATED CHANGE IN ELASTICITY AND TOTAL COST, BASE MODEL VERSUS NEUTRAL ADVERTISING, BY
DEMOGRAPHIC GROUP^a

Demographic Group	Mean Elasticity Base Model	Neutral Advertising	Neutral Advertising vs. Base Model		
			Percent Change in Total Cost	Average Savings Pesos per Capita (1997 Pesos)	Minimum Pension Equivalent per Capita (Months)
All	-0.754	-1.925	-17.2%	\$815	1.4
Low wage	-0.621	-0.997	-5.3%	\$96	0.2
High wage	-0.876	-2.779	-19.9%	\$1,475	2.5
Male	-0.797	-2.018	-17.8%	\$907	1.6
Female	-0.664	-1.728	-15.6%	\$620	1.1
Younger	-0.821	-1.677	-14.0%	\$521	0.9
Older	-0.680	-2.200	-19.5%	\$1,139	2.0

^aComputed using model estimates from equations (4)–(6), and estimated impact of sales force from Table III, column 4, and Table V, column 1. Pesos are reported in 1997 pesos. Minimum pension equivalents represent the “Average Savings in Pesos” as years of minimum pension guarantee payments. These payments are 580 pesos per month and are distributed to those who have less than a

6.7 Tables VII–IX: policy counterfactuals

Read:

- Table VII lets Afores reoptimize fees under neutral agents. Total costs fall by 61.6%, with large gains for every demographic group.
- Table VIII shows that a government competitor can fail when private firms respond discontinuously or when inelastic consumers remain captive.

- A government competitor alone can induce pricing at caps or generate limited savings if demand remains inelastic.
- Table IX combines demand-side and supply-side policies.
- Increasing price sensitivity for the most inelastic consumers lowers costs by 33.9%.
- Adding a government competitor raises this reduction to 38.5%.
- Combining stronger demand elasticity, a government competitor, and neutral agents lowers costs by 73.5%.

Economic message: Supply-side entry is not enough in a market with inelastic and captive consumers. The most effective policy package changes demand elasticity and the persuasive impact of sales force.

TABLE VII
SUMMARY OF OUTCOMES FOR DEMAND AND SUPPLY-SIDE POLICIES^a

	Simulated Outcomes From Neutral Agents' Preferences at New Equilibrium Fees		
	Percent Change in Total Cost	Average Savings Pesos per Capita (1997 Pesos)	Minimum Pension Equivalent per Capita (Months)
All	-61.6%	\$2,136	3.7
Low income	-54.9%	\$695	1.2
High income	-63.9%	\$3,455	6.0
Young workers	-63.0%	\$1,119	1.9
Old workers	-62.0%	\$3,257	5.6
Male	-59.8%	\$2,628	4.5
Female	-64.0%	\$1,901	3.3

^aEquilibrium calculations are based on an 80,229 random sample plus a proportional random sample of new workers who entered the market over time, to capture growth forecasts in market size. Equilibrium fees are calculated from an iterated best-response method using a 0.00025 grid for the base model while for the models with raised caps, equilibrium fees are calculated from an iterated best-response method using a 0.0005 grid. See the Supplemental Material for details on iterated best-response method. Cost is calculated over the whole account horizon and discounted at a 5% rate. A firm is at the cap if either equilibrium flow or balance fee is set

Table VII: demand- and supply-side policy outcomes

TABLE VIII
SUMMARY OF SIMULATION RESULTS WITH DISCONTINUOUS RESPONSES TO GOVERNMENT COMPETITOR^a

	Base Model With Gov't. Competitor		Neutral Agents With Gov't. Competitor
	(1) Cap at (2.00, 2.00)	(2) Cap at (3.20, 5.00)	(3) Cap at (3.20, 5.00)
Base Model	(1.90, 0.00)	(0.00, 0.10)	(0.00, 0.10)
XXI price	2.8%	5.8%	6.4%
XXI simulated share	-	94.0%	23.5%
Market share of Afores Pricing at cap	-	88.2%	20.7%
Percent of Afores pricing at cap	-	-	-
<i>Percentage change in management costs</i>			
All	-	-1.7%	7.6%
Low income	-	11.9%	20.0%
High income	-	-6.3%	2.2%
Male	-	-1.4%	7.0%
Female	-	-7.6%	0.8%
Young workers	-	5.9%	15.6%
Old workers	-	-9.4%	-1.9%

^aEquilibrium calculations are based on an 80,229 random sample plus a proportional random sample of new workers who entered the market over time, to capture growth forecasts in market size. Equilibrium fees are calculated from an iterated best-response method using a 0.00025 grid for the base model while for the models with raised caps, equilibrium fees are calculated from an iterated best-response method using a 0.0005 grid. See the Supplemental Material for details on iterated best-response method. Cost is calculated over the whole account horizon and discounted at a 5% rate. A firm is at the cap if either equilibrium flow or balance fee is set

Table VIII: discontinuous responses to government competitor

TABLE IX
SUMMARY OF OUTCOMES FOR DEMAND AND SUPPLY-SIDE POLICIES^a

	Policy Simulation		
	(1) Increased Demand Elasticity for Most Inelastic	(2) + Government Competitor	(3) + Neutral Agents
Market share of Afores pricing at cap	5.9%	0.0%	0.0%
Percent of Afores pricing at cap	10.3%	0.0%	0.0%
<i>Percentage change in management costs</i>			
All	-33.9%	-38.5%	-73.5%
Low income	-26.4%	-22.1%	-60.0%
High income	-36.7%	-44.5%	-78.0%
Young workers	-30.7%	-28.0%	-67.5%
Old workers	-37.8%	-49.5%	-79.9%
Male	-34.4%	-39.1%	-74.1%
Female	-36.2%	-44.6%	-76.7%

^aEquilibrium calculations are based on an 80,229 random sample plus a proportional random sample of new workers who entered the market over time, to capture growth forecasts in market size. Equilibrium fees are calculated from an iterated best-response method using a 0.00025 grid for the base model while for the models with raised caps, equilibrium fees are calculated from an iterated best-response method using a 0.0005 grid. See the Supplemental Material for details on iterated best-response method. Cost is calculated over the whole account horizon and discounted at a 5% rate. A firm is at the cap if either equilibrium flow or balance fee is set

7 Short summary

- Mexico's privatized pension system had many Afores but high fees.
- Afores set fees nationally but chose sales-force intensity locally.
- Large sales forces were associated with high fees and high market shares.
- Workers often chose Afores that were not the cheapest for them.
- The authors estimate a logit demand model with expected cost, brand value, and sales-force exposure.

- Sales force raises Afore brand value.
- Sales force also reduces price sensitivity.
- Neutralizing sales-force effects makes demand more elastic and lowers consumer costs.
- A government competitor alone may not discipline the market because inelastic consumers remain profitable to exploit.
- Policies that raise price sensitivity or reduce persuasive selling are necessary for robust competition.

8 Conclusion

8.1 Core conclusion

Hastings, Hortaçsu, and Syverson show that sales-force-based marketing can soften competition in financial product markets. In Mexico's privatized pension system, sales agents raised brand value and reduced consumer price sensitivity, helping high-fee Afores gain market share and sustain high equilibrium fees.

8.2 Economic intuition

If consumers are financially inattentive or face complex fees, firms may find it more profitable to invest in persuasive marketing than in price cuts. Sales agents can make a product salient, create trust, and steer clients toward a provider. When consumers become less responsive to costs, firms have weaker incentives to lower fees.

8.3 Incremental contribution takeaway

The paper's incremental contribution is to connect micro-level sales-force deployment to equilibrium competition and welfare in a large real-world financial market. It moves beyond the statement that consumers make mistakes and shows how firms strategically exploit and amplify those mistakes through nonprice competition.

8.4 Policy takeaway

Regulators cannot rely only on supply-side competition when consumers are inelastic. A low-cost government competitor or more firms may be insufficient. Effective policy must also address demand-side frictions: fee salience, financial literacy, switching costs, and the persuasive effects of sales agents.